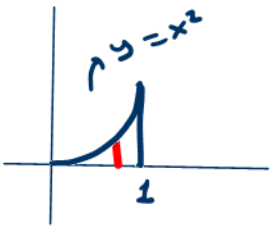
Cevaplar

① $\int_0^1 \int_{\sqrt{y}}^1 \frac{dx\,dy}{\sqrt{1+3x^3}}$ integrasyon sırasını değiştirirsek

D: $y=1$ $y=0$ → x-eks.
$x=\sqrt{y}$ $x=1$
↓
$y=x^2$



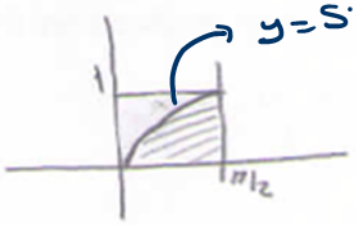
$I = \int_0^1 \int_0^{x^2} \frac{dy\,dx}{\sqrt{1+3x^3}}$

Cevap B

---

② $\int_0^1 \int_{\text{Arcsiny}}^{\pi/2} e^{\cos x}\,dx\,dy$ int. sırasını değiştirirsek yeni integral?

D: $y=1$ $y=0$
$x=\frac{\pi}{2}$ $x=\text{ArcSiny}$
$y=\text{Sinx}$



$I = \int_0^{\pi/2} \int_0^{\text{Sinx}} e^{\cos x}\,dy\,dx = \int_0^{\pi/2} \int_0^{\text{Sinx}} e^{\cos x}\,dy\,dx$

Cevap C

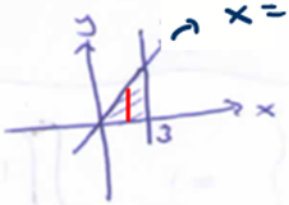
③

$$\int_0^1 \int_{3y}^{3} e^{x^2}\,dx\,dy$$ int. sirasini degistirince yeni integral?

a) $\int_0^3 \int_0^{x/3} e^{x^2}\,dx\,dy$ b) $\int_0^3 \int_0^{x/3} e^{x^2}\,dy\,dx$ c) $\int_{3y}^{3} \int_0^1 e^{x^2}\,dy\,dx$

x-ekseni

O: $y=0$ $y=1$ $x=3$ $x=3y$



$x=3y \rightarrow y=\frac{x}{3}$

$$I=\int_0^3 \int_0^{x/3} e^{x^2}\,dy\,dx$$

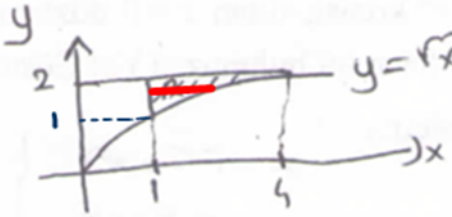
Cevap B

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④

$$\int_1^4 \int_{\sqrt{x}}^{2} \frac{e^y}{y+1}\,dy\,dx$$ int. sirasini degistirince yeni int.?

R: $\begin{cases} y=\sqrt{x}, & y=2 \\ x=1, & x=4 \end{cases}$



$y=\sqrt{x} \rightarrow x=y^2$

$$I=\int_1^2 \int_1^{y^2} \frac{e^y}{y+1}\,dx\,dy$$

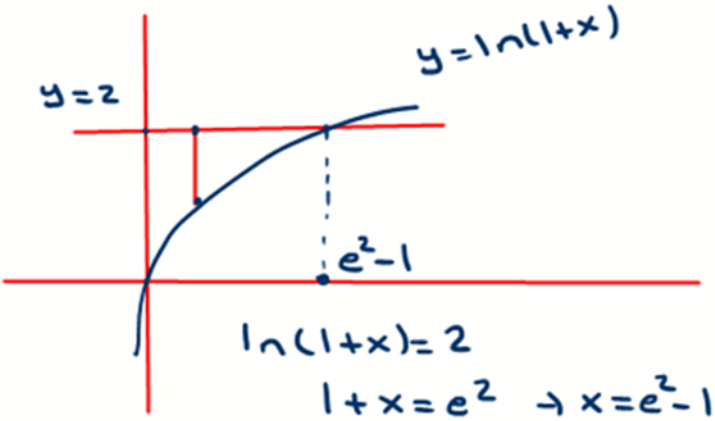
Cevap B

(5)

$y=\ln(1+x)$, $x=0$, $y=2$ ile sınırlı bölgenin alanını veren integral hangisi(leri) olabilir?

I. $\int_0^2 \int_0^{e^y} dx\,dy$ II. $\int_0^{e^2-1} \int_{\ln(1+x)}^2 dy\,dx$ III. $\int_0^2 \int_0^{e^y-1} dx\,dy$

a) I, II, III b) I, II c) II, III d) II e) III

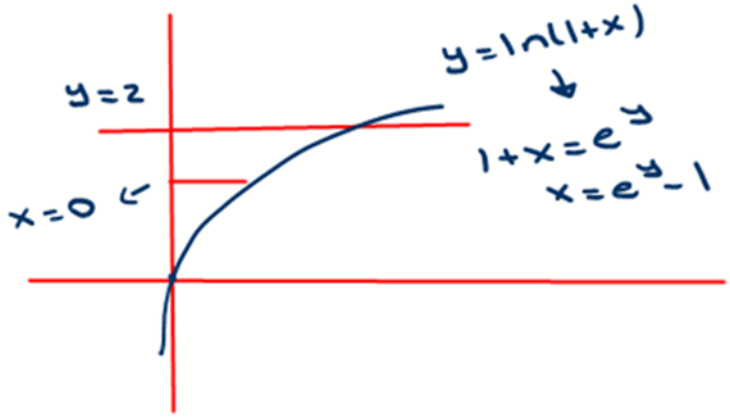


$\ln(1+x)=2$

$1+x=e^2 \rightarrow x=e^2-1$

★ x'e göre

$$\int_0^{e^2-1} \int_{\ln(1+x)}^2 dy\,dx \rightarrow \text{III}$$



$1+x=e^y$

$x=e^y-1$

★ y'ye göre

$$\int_0^2 \int_0^{e^y-1} dx\,dy \rightarrow \text{III}$$

Cevap C

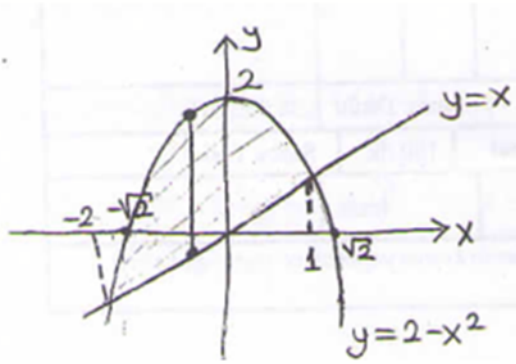
⑥

D bölgesi $y = 2-x^2$ ve $y = x$ ile sınırlı bölge olsun. $z = 2x^2y$ ile üstten, $z=0$ daki D bölgesi ile alttan sınırlanan cismin hacmini veren integral?

a) $\int_0^1 \int_x^{2-x^2} 2x^2y \, dy\,dx$

b) $\int_{-2}^1 \int_x^{2-x^2} 2x^2 \, dy\,dx$

c) $\int_{-2}^2 \int_0^{2-x^2} 2x^2y \, dy\,dx$



$2-x^2 = x$

$x^2+x-2 = 0$

$x=-2, \; x=1$

$$V = \iint_D 2x^2y \, dx\,dy$$

$$= \int_{-2}^1 \int_x^{2-x^2} 2x^2y \, dy\,dx$$

Cevap B

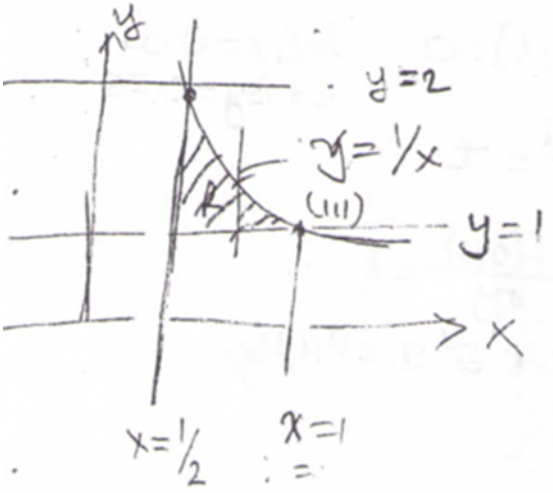
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⑦ R: $y=x$, $y=x^3$, $x \geq 0$, $y \geq 0$ olsun. $f(x,y) = e^{2x^2-x^4}$ fonksiyonunun R bölgesindeki iki katlı int. değeri?



$$\iint_R e^{2x^2-x^4} \, dA = \int_0^1 \int_{x^2}^{x} e^{2x^2-x^4} \, dy\,dx = \int_0^1 e^{2x^2-x^4}(x-x^3)\,dx = \int_0^1 \frac{1}{4} e^u \, du = \frac{1}{4}(e-1)$$

⑧ $\int_{1}^{2}\int_{1/2}^{1/y} e^{\ln x - x}\,dx\,dy = ?$



$$I = \int_{x=1/2}^{1}\left(\int_{y=1}^{1/x} e^{\ln x - x}\,dy\right)dx$$

$$= \int_{1/2}^{1} e^{\ln x - x}\, y\Big|_{1}^{1/x}\,dx$$

$$= \int_{1/2}^{1}\left(\frac{1}{x}-1\right)e^{\ln x - x}\,dx = e^{\ln x - x}\Big|_{1/2}^{1} = 1\cdot e^{-1} - \frac{1}{2}e^{-1/2}$$

$$= \frac{1}{e} - \frac{1}{2\sqrt{e}} //$$

---

⑨ $z = 1+2xy$ yüzeyinin altında; $y=x^2$, $y=2-x^2$ paraboller ile sınırlı bölgenin üstünde kalan T cisminin hacmi?



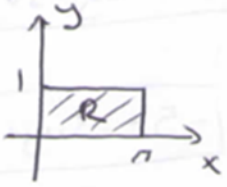
$y=x^2$, $y=2-x^2$ $\Rightarrow$ $x^2 = 2-x^2 \Rightarrow x = \pm 1$

$$V = \iint_{D}(1+2xy)\,dx\,dy = \int_{-1}^{1}\int_{x^2}^{2-x^2}(1+2xy)\,dy\,dx$$

$$= \int_{-1}^{1} y + xy^2\Big|_{x^2}^{2-x^2}\,dx$$

$$= \int_{-1}^{1}\left(2-x^2 + x(2-x^2)^2 - x^2 - x\cdot x^4\right)dx = \frac{8}{3}$$

(10) $f(x,y) = x\cos(xy)$ fonksiyonunun $R: \begin{cases} 0 \le x \le \pi \\ 0 \le y \le 1 \end{cases}$ bölgesindeki iki katlı integralinin değeri?



$$\iint_R f(x,y)\,dA = \int_0^{\pi}\int_0^{1} x\cos xy\; dy\,dx$$

$$= \int_0^{\pi} \sin xy\,\Big|_0^1 dx = \int_0^{\pi} \sin x\,dx = (-\cos x)\Big|_0^{\pi} = -(-1-1) = 2$$

---

(11) $$\int_1^4 \int_{\sqrt{x}}^2 \frac{e^y}{y+1}\,dy\,dx = ?$$



$$R: \begin{cases} y = \sqrt{x}, & y = 2 \\ x = 1, & x = 4 \end{cases}$$

$$R = \{(x,y) : 1 \le x \le 4,\ \sqrt{x} \le y \le 2\}$$

$$R' = \{(x,y) : 1 \le y \le 2,\ 1 \le x \le y^2\}$$

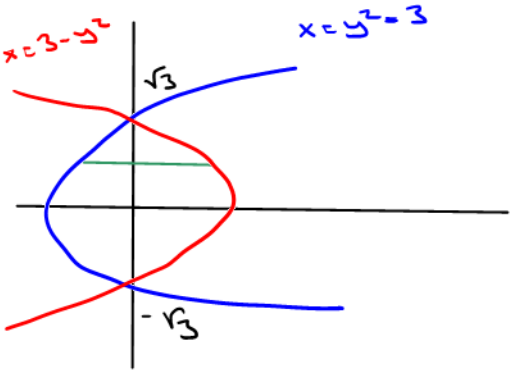
$$I = \int_1^4 \int_{\sqrt{x}}^2 \frac{e^y}{y+1}\,dy\,dx = \int_1^2 \int_1^{y^2} \frac{e^y}{y+1}\,dx\,dy = \int_1^2 \frac{e^y}{y+1}(y^2-1)\,dy$$

$$= \int_1^2 (y-1)e^y\,dy = (y-2)e^y\Big|_1^2 = e$$

$$\begin{bmatrix} u = y-1 \\ dv = e^y dy \end{bmatrix} \to du = dy,\ v = e^y \Rightarrow uv - \int v\,du = (y-1)e^y - \int e^y dy = (y-2)e^y$$

(12) $x = 3 - y^2$ ve $x = y^2 - 3$ eğrileri ile sınırlı düzlem bölgenin alanını veren integral aşağıdakilerden hangisidir?

A) $\int_0^3 \int_{3-y^2}^{y^2-3} dx\,dy$ B) $\int_{-\sqrt{3}}^{\sqrt{3}} \int_{3-y^2}^{y^2-3} dy\,dx$ C) $\int_{-\sqrt{3}}^{\sqrt{3}} \int_{3-y^2}^{y^2-3} dx\,dy$ D) $\int_{-\sqrt{3}}^{\sqrt{3}} \int_{y^2-3}^{3-y^2} dx\,dy$ E) $\int_{-\sqrt{3}}^{\sqrt{3}} \int_{y^2-3}^{3-y^2} dy\,dx$



$$I = \int_{-\sqrt{3}}^{\sqrt{3}} \int_{y^2-3}^{3-y^2} dx\,dy$$

---

(13) $\int_0^{\frac{3}{4}} \int_{3x}^{3-x} f(x,y)\,dy\,dx$ integralini integrasyon sırasını değiştirerek yeniden yazdığımızda elde edilen yeni integral aşağıdakilerden hangisidir?

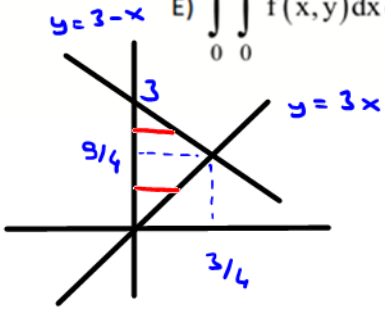
A) $\int_{3x}^{3-x} \int_0^{\frac{3}{4}} f(x,y)\,dx\,dy$ B) $\int_0^{\frac{3}{4}} \int_0^{\frac{y}{3}} f(x,y)\,dx\,dy + \int_{\frac{3}{4}}^{3} \int_{\frac{y}{3}}^{3-y} f(x,y)\,dx\,dy$ C) $\int_0^{\frac{9}{4}} \int_0^{\frac{y}{3}} f(x,y)\,dx\,dy + \int_{\frac{9}{4}}^{3} \int_{\frac{y}{3}}^{3-y} f(x,y)\,dx\,dy$

D) $\int_0^{\frac{9}{4}} \int_0^{\frac{y}{3}} f(x,y)\,dx\,dy + \int_{\frac{9}{4}}^{3} \int_0^{3-y} f(x,y)\,dx\,dy$ E) $\int_0^{\frac{3}{4}} \int_0^{\frac{y}{3}} f(x,y)\,dx\,dy + \int_0^{\frac{9}{4}} \int_0^{3-y} f(x,y)\,dx\,dy$

y - ekseni

$x = \frac{3}{4}$ $x = 0$

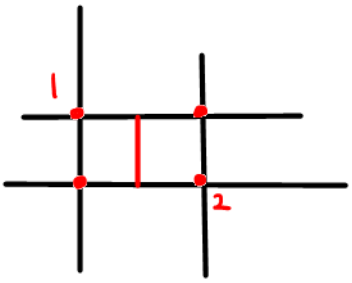
$y = 3 - x$ $y = 3x$



$$I = \int_0^{9/4} \int_0^{y/3} f\,dx\,dy + \int_{9/4}^{3} \int_0^{3-y} f\,dx\,dy$$

$x=0 \longleftrightarrow x = y/3$ $\qquad$ $x=0 \longleftrightarrow x = 3-y$

(14) Köşeleri $(0,0),(0,1),(2,0)$ ve $(2,1)$ noktalarında bulunan dikdörtgensel bir bölgede $f(x,y)=xy+2y$ fonksiyonunun iki katlı integralinin değeri?



$$I=\iint_D (xy+2y)\,dy\,dx$$

$$=\int_0^2\int_0^1 (xy+2y)\,dy\,dx=\int_0^2 \left(x\frac{y^2}{2}+y^2\right)\Big|_0^1 dx$$

$$=\int_0^2 \left(\frac{x}{2}+1\right)dx=\frac{x^2}{4}+x\Big|_0^2$$

$$=1+2=3$$

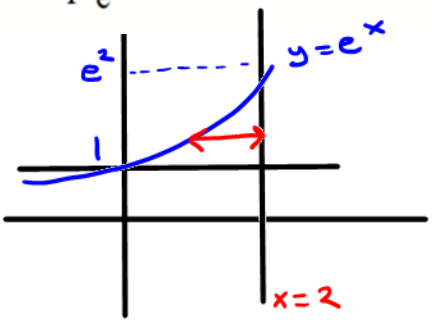
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(15) $\int_0^2\int_1^{e^x} \sin x^3\,dy\,dx$ integralini İntegrasyon sırasını değiştirerek

D: $x=2$, $x=0$, $y=e^x$, $y=1$

yeniden yazdığımızda elde edeceğimiz integral aşağıdakilerden hangisidir?

(a) $\int_1^{e^2}\int_{e^x}^2 \sin x^3\,dx\,dy$ (b) $\int_1^{e^2}\int_{\ln y}^2 \sin x^3\,dx\,dy$ (c) $\int_1^2\int_{e^x}^2 \sin x^3\,dx\,dy$ (d) $\int_1^2\int_{\ln y}^2 \sin x^3\,dx\,dy$ (e) $\int_1^{e^x}\int_0^2 \sin x^3\,dx\,dy$



$\longleftrightarrow$ $x=2$

$y=e^x$

$\downarrow$

$x=\ln y$

$$I=\int_1^{e^2}\int_{\ln y}^2 \sin x^3\,dx\,dy$$

(16) D bölgesi: $y=\sqrt{x}$ eğrisi ile $y+x=2$ ve $y=0$ doğruları ile sınırlı bölge olmak üzere, D bölgesinin alanını veren iki katlı integral aşağıdakilerden hangisidir?

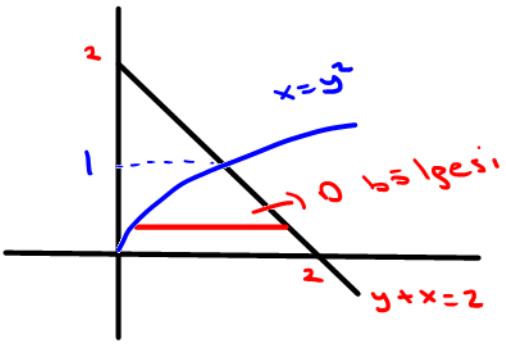
(a) $\int_0^1 \int_{\sqrt{x}}^{2-x} dy\,dx$

(b) $\int_0^1 \int_{y^2}^{2-y} dx\,dy$

(c) $\int_0^1 \int_{x^2}^{2-x} dx\,dy$

(d) $\int_0^1 \int_{\sqrt{y}}^{2-y} dy\,dx$

(e) $\int_0^1 \int_{2-y}^{y^2} dx\,dy$



x'e göre düzgün — 2 integral

y'ye göre düzgün — 1 integral

* şıklarda tek integral olduğun- dan y'y göre düzgün bölge olarak çözmeliyiz.

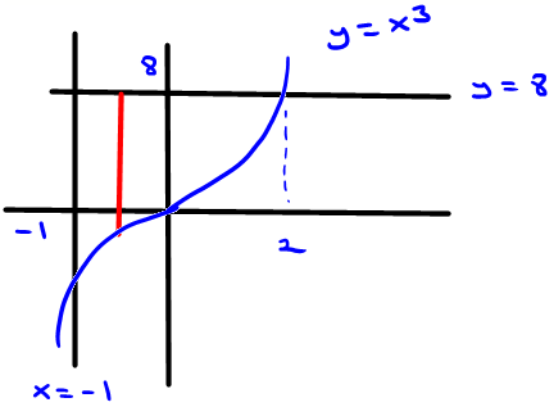
$$A=\int_0^1 \int_{y^2}^{2-y} dx\,dy$$

$x=y^2 \longleftrightarrow x=2-y$

---

(17) D bölgesi: $y=x^3$ eğrisi ile $y=8$ ve $x=-1$ doğruları ile sınırlı bölge olmak üzere, D bölgesinin alanını veren iki katlı integral aşağıdakilerden hangisidir?

A) $\int_{-1}^{2} \int_{x^3}^{-1} dy\,dx$ B) $\int_{-1}^{2} \int_{-8}^{x^3} dy\,dx$ C) $\int_{-1}^{2} \int_{\sqrt[3]{y}}^{8} dy\,dx$ D) $\int_{-1}^{2} \int_{x^3}^{8} dy\,dx$ E) $\int_{-1}^{2} \int_{x^3}^{0} dy\,dx$



$$A=\int_{-1}^{2} \int_{x^3}^{8} dy\,dx$$

$y=x \quad y=\frac{x^2}{9} \qquad y=1 \quad y=\frac{x^2}{9}$

$x=1 \quad x=0 \qquad x=1 \quad x=3$

(18) $\int_0^1 \int_{\frac{x^2}{9}}^{x} f(x,y)\,dy\,dx + \int_1^3 \int_{\frac{x^2}{9}}^{1} f(x,y)\,dy\,dx$ integralini integrasyon sırasını değiştirerek yeniden yazdığımızda elde edilen yeni integral aşağıdakilerden hangisidir?

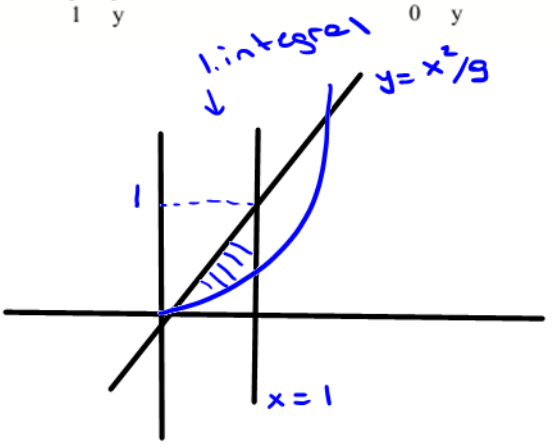
A) $\int_0^1 \int_{\frac{y^2}{9}}^{y} f(x,y)\,dx\,dy + \int_1^3 \int_{\frac{y^2}{9}}^{1} f(x,y)\,dx\,dy$

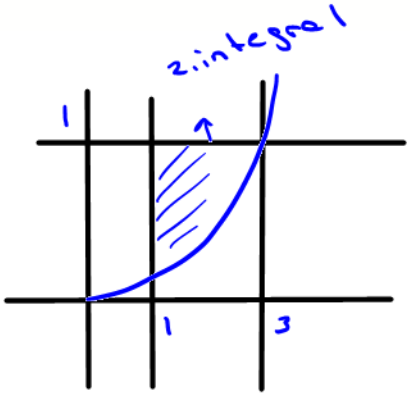
B) $\int_0^1 \int_y^{3\sqrt{y}} f(x,y)\,dx\,dy$

C) $\int_0^1 \int_{3\sqrt{y}}^{y} f(x,y)\,dx\,dy$

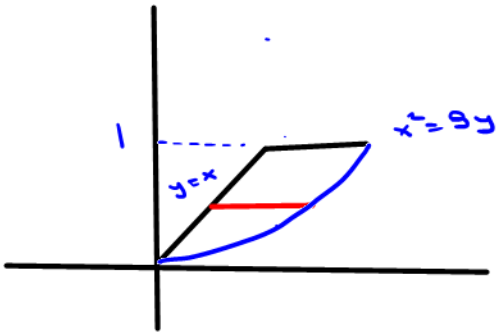
D) $\int_1^3 \int_y^{3\sqrt{y}} f(x,y)\,dx\,dy$

E) $\int_0^1 \int_y^{\frac{y^2}{9}} f(x,y)\,dx\,dy + \int_1^3 \int_1^{\frac{y^2}{9}} f(x,y)\,dx\,dy$





birleştirelim



$$I = \int_0^1 \int_y^{3\sqrt{y}} f\,dx\,dy$$

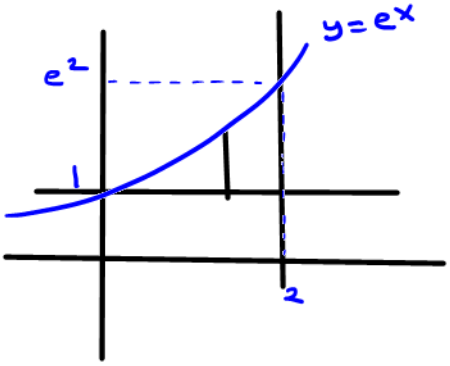
$x=y \quad \longleftrightarrow \quad x=3\sqrt{y}$

D: $y=1$ $y=e^2$ $x=2$ $x=\ln y$ → $y=e^x$

(19) $\int_1^{e^2}\int_{\ln y}^{2} f(x,y)dxdy$ integralini integrasyon sırasını değiştirerek yeniden yazdığımızda elde edilen yeni integral aşağıdakilerden hangisidir?

A) $\int_0^2\int_1^{e^x} f(x,y)dydx$ B) $\int_{\ln y}^2\int_1^{e^2} f(x,y)dydx$ C) $\int_0^2\int_0^{e^x} f(x,y)dydx$

D) $\int_1^{e^2}\int_{e^x}^2 f(x,y)dydx$ E) $\int_1^{e^2}\int_{e^x}^1 f(x,y)dydx$



$$I=\int_0^2\int_1^{e^x} f\,dy\,dx$$